

Arjun Sood

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Education

Georgia Institute of Technology | Atlanta, GA

Expected Graduation, May 2028

Bachelor of Science in Computer Science, Concentrations in AI + Modeling & Simulation, GPA 3.55/4.00

Experience

Georgia Institute of Technology RICAL Robotics Laboratory | Atlanta, GA

August 2025 - Present

Robotics Engineer & Researcher

- Developed RL training pipelines in NVIDIA Isaac Lab for Unitree Go2 locomotion - reward shaping, hyperparameter tuning, curriculum learning across 4,096 GPU-parallel environments and 17 reward terms - improving rough-terrain success from ~62% to 89%.
- Created a reusable ROS 2 package with approximately 3 modular nodes for command input, wheel control, and system telemetry to interface wheel controls and make the platform easy to integrate into other robotics projects.

Color Robotics | San Francisco, CA

March 2025 - October 2025

Software Engineering Intern

- Designed an AI-powered diagnostic agent (LLMs, RAG, knowledge graphs) supporting a 5-stage workflow: issue analysis, maintenance-data synthesis, solution recommendation, and resolution reporting.
- Developed webpages and elements using HTML, CSS, React JS, Python, and Django for a robot monitoring platform displaying 12+ categories of live backend performance, metrics, and deployment data.

University of Pennsylvania ASSET Laboratory | Philadelphia, PA

June 2024 - August 2025

AI/ML Engineer & Researcher

- Researched, developed, and implemented a selective LLM fine-tuning and quantization technique in PyTorch and Python, achieving model performance within approximately 2 percentage points of full fine-tuning despite reducing non-zero bits by 50%.
- Co-led a project training an LLM in PyTorch using Group Relative Policy Optimization to generate Python scripts that search medical databases and predict patient outcomes under two treatment conditions.

Projects [\(Portfolio\)](#)

Quant Research Platform & C++ Event-Driven Backtesting Engine | C++20, DuckDB, Parquet, pybind11, Python | [Repo Link](#)

- Wrote a C++20 limit-order-book replay engine with zero-copy pybind11/NumPy bindings (GIL released in the hot loop), processing 11.5M events/sec - ~488x the pure-Python reference - with bit-identical cross-language parity enforced in CI.
- Built a market-data research platform ingesting L2 order-book/trade feeds into hive-partitioned Parquet, achieving 2.5-2.9x faster DuckDB queries than a pandas baseline via partition pruning and predicate pushdown.

Raft-Replicated KV Store: LSM Engine & Deterministic Simulation Testing | Rust, Tokio | [Repo Link](#)

- Built a linearizable distributed KV database from scratch (no DB/consensus/RPC libraries): LSM-tree storage engine with CRC-framed WAL, bloom-filtered SSTables, and MVCC snapshots, under a full Raft implementation.
- Engineered a deterministic simulator executing 20,000 failure schedules (~420M events), catching 8 real bugs pre-release; shipped a tokio production server with group commit and three read-consistency modes.

LLM Pretraining, Post-Training, and High-Performance Inference Engine | PyTorch, Triton, FSDP2 | [Repo Link](#)

- Built an end-to-end LLM platform - from-scratch BPE tokenizer, LLaMA-style transformer (RoPE/GQA/SwiGLU), and a DDP/FSDP2 pretraining stack with custom Triton kernels - training a 125M model at 42.7% MFU on a single RTX 5070.
- Engineered a vLLM-style inference engine: paged KV cache with prefix caching, continuous batching with recompute preemption, speculative decoding (3.09 output tokens per target forward at 53% draft acceptance), and int8 weight-only quantization.

Skills

Systems & Distributed Computing: Raft Consensus, LSM Trees, WAL/SSTables, MVCC, Bloom Filters, Deterministic Simulation, Async Programming, Tokio, Multithreading, WebSockets, MessagePack

Languages: C/C++ (C++20), Python, Rust, Java, SQL, Bash, JavaScript

Data & Quant Infrastructure: DuckDB, Parquet, L2 Order Books, Event-Driven Backtesting, Market Microstructure Features, FastAPI, Prometheus

Development & Tools: Git, Linux, Docker, GitHub Actions, CI/CD, pytest, CMake